

Open-Qwen-Music: An Auditable Framework for LLM-Based Music Composition and Diffusion Rendering

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Abstract

We present Open-Qwen-Music, an open reconstruction of Qwen-Music and a fully specified research system for text-to-music generation that couples LLM-based semantic composition with diffusion-based acoustic rendering. The system comprises a 25 Hz single-codebook music tokenizer, a 3B-parameter autoregressive Music LLM, and a diffusion renderer producing 48 kHz stereo audio, following the cross-module interfaces reported by Qwen-Music. The strongest systems of this design remain closed, and prominent open music-generation projects release weights and inference code without their training corpora or end-to-end training implementations. This limits independent and controlled study of how information loss and prediction errors propagate from semantic representation through autoregressive planning to acoustic rendering. To our knowledge, Open-Qwen-Music is the first fully open release of an LLM-composition-plus-diffusion-rendering text-to-music system. Beyond model weights and inference code, the release includes the training datasets and provenance manifests, complete data-processing, annotation, training, inference, and evaluation pipelines, configurations, and pretrained weights for every learned module. Artifact manifests bind the identities of these artifacts across the complete workflow. Together, these artifacts establish a reproducible implementation of the modular architecture and provide an empirical basis for component-level analysis and future evaluation. We present the system as a transparent, executable research baseline and a starting point for the community, not as evidence of quality parity with Qwen-Music. Open-Qwen-Music is an ongoing effort, and we will continue to improve its generation quality, controllability, and robustness. All release artifacts are available at <https://github.com/biang15343100-source/Open-Qwen-Music>.

1 Introduction

Recent music-generation systems increasingly separate semantic composition from acoustic rendering. A language model first plans long-range musical content in a compact representation. A diffusion or flow model then renders that representation into audio. Qwen-Music, ACE-Step 1.5, and LeVo 2 all follow this broad design (Xu et al., 2026; Gong et al., 2026; Lei et al., 2026). This decomposition offers a natural way to divide musical planning from waveform synthesis, but it also creates coupled failure modes: information discarded by the representation constrains the planner, and planner errors become out-of-distribution conditions for the renderer. Studying these interactions requires more than inference access to a pretrained model.

Many prominent open music-generation projects release pretrained weights and inference code but not the original training corpora or complete training implementations (Yuan et al., 2026; Yang et al., 2026; Lei et al., 2026; Zhang et al., 2025; Yang et al., 2025). Fully open alternatives demonstrate the value of releasing data and training code, but may adopt architectures without a separately trained diffusion renderer (Jiang et al., 2026). Consequently, the effects of changing one module on the rest of an autoregressive composition and diffusion-rendering pipeline remain difficult to measure.

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Qwen-Music combines a 25 Hz single-codebook music tokenizer, an autoregressive Music LLM, and a diffusion-based renderer (Xu et al., 2026). Its authors report a 50.3% preference rate against Suno V5.5 in a private bilingual blind test. The Qwen-Music report describes the high-level architecture and staged training process. It does not release the training data, code, weights, or many details needed for exact reproduction. Missing details include corpus construction, token registration, optimization settings, sampling mixtures, and checkpoint selection. Reconstructing the system therefore requires both engineering choices and empirical validation.

We present Open-Qwen-Music, an open reconstruction of Qwen-Music and a fully open research framework for this modular paradigm. It combines a low-frame-rate semantic tokenizer, an autoregressive planner, and a diffusion-based acoustic renderer under explicit cross-module contracts: 25 Hz single-codebook Semantic Tokens, 6.25 Hz Melody Tokens, and 48 kHz stereo output. The framework is trained with publicly obtainable and locally generated data, including a synthetic corpus produced with an open-weight MiniMax Music 3 model (MiniMax AI, 2026). Beyond the reconstruction itself, the executable pipeline makes the interfaces among semantic representation, autoregressive generation, and acoustic rendering directly inspectable.

We ask two questions. First, how can a modular semantic-composition and acoustic-rendering system be trained under explicit, independently reproducible interfaces? Second, how can component-level diagnostics and artifact provenance make the behavior and limitations of a modular text-to-music system reproducible and inspectable?

Our main contributions are:

- We construct a complete modular system with explicit representation contracts, making the tokenizer, planner, spectral decoder, diffusion renderer, and refinement module independently trainable and replaceable.
- We report fixed-protocol evaluations of automatic supervision and semantic discretization, qualitative end-to-end observations, and auditable intermediate artifacts that support reproducible analysis of the released system. The reported measurements characterize this implementation rather than general properties of the architecture.
- We release the training datasets and provenance manifests, data-processing and annotation pipelines, complete training and inference implementations, evaluation code, configurations, and pretrained weights for every learned component at <https://github.com/biang15343100-source/Open-Qwen-Music>.

2 Dataset and Data Pipeline

We organize the data into three nested corpora and one independent synthetic set. Dataset-Inventory is the broad audio inventory. Dataset-Annotated is a deterministic, source-capped subset of Dataset-Inventory. Dataset-Annotation-Filtered is an annotation-filtered subset of Dataset-Annotated. MiniMax is generated with MiniMax Music 3 (MiniMax AI, 2026). Table 1 gives their scale and use.

Table 1: Dataset summary. Dataset-Annotation-Filtered is contained in Dataset-Annotated and is not counted as additional audio when both are used.

Dataset	Audio records	Audio hours	Definition and main use
Dataset-Inventory	1,481,301	63,170.407	Broad 26-source inventory for tokenizer and Spec-VAE training.
Dataset-Annotated	240,325	9,506.162	Source-capped, annotated subset of Dataset-Inventory.
Dataset-Annotation-Filtered	44,355	2,155.176	Strict annotation-filtered subset of Dataset-Annotated: 35,484 vocal and 8,871 instrumental records.
MiniMax	25,606	1,103.057	Independent synthetic augmentation: 10,197 Chinese, 9,955 English, and 5,454 instrumental records.
Research package	62,417	2,958.571	Access-screened Dataset-Annotation-Filtered audio plus MiniMax audio; 631 additional metadata-only records are not included in the audio count.

2.1 Corpus Construction

Dataset-Inventory. Dataset-Inventory indexes 26 heterogeneous research, public-download, and synthetic collections. Because these sources originate from independent open-data efforts, they carry no unified annotation schema—lyrics, section boundaries, and musical tags are absent, inconsistent, or source-specific. Records are variable-length source items; each model applies its own eligibility rules and sampler. Appendix A.1 lists every source, citation, and mirror or derivative caveat.

Dataset-Annotated. To enable the training of the Music LLM and the acoustic renderer, we developed a unified annotation pipeline that produces consistent lyrics, section structure, and musical tags across all sources. Annotating the entirety of Dataset-Inventory would be prohibitively expensive, so we cap each source at 20,000 records. The resulting corpus is Dataset-Annotated.

Dataset-Annotation-Filtered. Dataset-Annotation-Filtered is derived from Dataset-Annotated by applying strict annotation-quality filters and distributional balancing. On the quality side, vocal records require usable Melody-CoT labels, multiple sections, and reliable lyrics and tags; instrumental records must be non-synthetic and pass lyric, vocal-tag, and vocal-energy checks. Appendix A.2 summarizes the main rules. On the balancing side, a deterministic weighted selection is followed by mixed-integer optimization that enforces a 4:1 vocal-to-instrumental ratio (consistent with Qwen-Music) and limits concentration by source, gender, language, and generator. The result contains 26,044 synthetic vocal, 9,440 non-synthetic vocal, and 8,871 non-synthetic instrumental records. It is drawn only from Dataset-Annotated’s training split and has no separate validation or test split.

MiniMax generated set. We generate candidates with deterministic seeds, five target durations, and at most five retries. Quality checks cover duration, loudness, clipping, silence, DC offset, bandwidth, and spectral flatness. The retained audio is 44.1 kHz, 16-bit PCM stereo. Generation conditions follow a controlled taxonomy. `gemini-3.1-pro-preview` and `deepseek-v4-pro` provide lyrics for the generation requests; these are not treated as transcripts. The generated audio instead passes through the same ASR, alignment, and sectioning pipeline as the other corpora; only style conditions are retained directly.

Training views. Dataset-Inventory supplies the broad candidate pool for the tokenizer and Spec-VAE.

The Music LLM training corpus contains 291,569 English section-window records. Songs are segmented using their annotated sections, with every training window limited to 30 seconds, or 750 Semantic frames at 25 Hz. Its strict statistics entry point recomputes 25-Hz training exposure and overlap-merged canonical audio spans from that exact manifest.

2.2 Preprocessing and Annotation

Preprocessing validates source audio before deduplication and publication. Annotation separates stems, infers structure, cross-checks two ASR transcripts, aligns sections, estimates audio features and tags, and emits eligibility flags. Only samples that pass validation enter downstream corpora.

Preprocessing. The preprocessing pipeline is index-only: it does not resample or copy source audio. It records duration, sample rate, channel count, and eight quality indicators. It rejects invalid durations, sample rates below 24 kHz, bitrates below 64 kbps, fake stereo, severe bandwidth truncation, heavy boundary silence, invalid channels, and extreme peaks. The accepted duration range is 5–360 seconds. Deduplication proceeds through URI, byte hash, external identifier, and decoded-PCM fingerprint matching. These steps reduce 1,823,837 accepted parent records, totaling 83,022.76 hours, to Dataset-Inventory.

Annotation. The annotation pipeline has 12 ordered stages. We separate vocals and accompaniment with Hybrid Demucs (Défossez, 2021); the backend is `HDEMUCS_HIGH_MUSDB_PLUS`. SongFormer proposes song sections (Hao et al., 2025). Qwen3-ASR-1.7B transcribes both the vocal stem and the mixture, and Qwen3-ForcedAligner-0.6B assigns timestamps (Shi et al., 2026). We cross-check both transcripts before section fusion. Recent work on recognition under compound real-world distortions shows that weakened acoustic

grounding can produce omissions and hallucinations (Xie et al., 2026b); accordingly, both transcription paths are treated as uncertain supervision rather than human ground truth.

We then estimate voice activity, pitch, and spectral properties. pYIN is used for annotation-time pitch analysis (Mauch & Dixon, 2014); the separate Melody Tokenizer uses RMVPE, as described in Section 3.2. A controlled Qwen3-Omni-30B-A3B-Instruct tagger produces genre, mood, instrument, vocal-gender, and vocal-timbre labels plus a short description (Xu et al., 2025b). The final stage combines this evidence into task-specific eligibility flags.

Annotation limits. All annotations—lyrics, sections, tags, descriptions—are produced by off-the-shelf open-source models used without any training or fine-tuning. They are therefore automatic estimates, not human ground truth, and may contain systematic or per-sample errors. Confidence thresholds serve only as filters, not accuracy estimates. Dataset-Annotation-Filtered is not an acoustic-fidelity set, so Render uses a separate acoustic gate.

2.3 Research Data Release

To let the community build on our work without repeating the costly annotation pipeline, we publicly release the two highest-quality corpora from our pipeline—Dataset-Annotation-Filtered and the MiniMax generated set—after applying a permission-aware desensitization process. The pinned research package is available at <https://huggingface.co/datasets/david-miller-45678/open-qwen-music-dataset-c-minimax-music3>. It combines 36,811 access-screened Dataset-Annotation-Filtered audio records with all 25,606 MiniMax records. We exclude 6,912 records from the Jamendo-QA mirror (Archit00, 2026; Koh et al., 2025), MoisesDB (Pereira et al., 2023), MuChin-v2 (karl-wang, 2025; Wang et al., 2024), and Music4All (Santana et al., 2020) because those sources require permission for access or redistribution. Cambridge-MT (Senior, n.d.) contributes 631 metadata-only records. One JamendoMaxCaps record is quarantined by the privacy scanner. The artifact manifest records source-level access and redistribution decisions separately for audio, annotations, tokens, latents, and model weights, and the public package contains only artifacts covered by the corresponding release policy.

3 Model and Training

The system reconstructs Qwen-Music’s tokenizer, autoregressive Music LLM, and diffusion Render modules (Xu et al., 2026). Table 2 fixes their shared interfaces; changing any entry invalidates downstream assets and checkpoints.

Table 2: Cross-module interfaces.

Representation	Rate	Specification	Downstream role
Semantic Tokens	25 Hz	one codebook, 32,768	Music LLM and DiT
Melody Tokens	6.25 Hz	256; unvoiced = 255	Melody-CoT
Spectral latent	25 Hz	128 dimensions	Spec Decoder
Waveform	48 kHz	stereo	playback and evaluation

3.1 Music Tokenizer

Architecture. The tokenizer maps 24 kHz mono audio to a 25 Hz stream from one 32,768-entry codebook, a nominal rate of 375 bit/s. A 128-bin log-Mel frontend ($n_{\text{fft}} = 2048$, hop 240, $f_{\text{max}} = 12$ kHz) runs at 100 Hz. A causal ConvNeXt frontend (kernel 7, expansion 4) reduces the rate by four before a 24-layer, 1024-wide Conformer with 16 attention heads, a 4096-dim feed-forward network, and RoPE. A single 16-dimensional cosine vector quantizer is inserted after layer 13; three lightweight heads predict multilingual CTC subwords, log-Mel, and soft Chroma. Deployment retains only the causal encoder up to the quantizer. The frontend and all Conformer convolutions are causal in every stage; self-attention becomes strictly causal during the causal-adaptation and subsequent phases.

Qwen-Music specifies the broad architecture and four training stages but omits most feature, optimization, and budget settings. The choices below are our reconstruction decisions, not claims about the official implementation.

Tokenizer training recipe. Table 3 summarizes the five functional phases used to train the semantic tokenizer.

Table 3: Training recipe for the semantic tokenizer.

Phase	Objective	Main reconstruction choice
Pretraining	Bidirectional BestRQ	Broad pretraining with waveform masking and four-frame targets.
Domain adaptation	BestRQ	Fresh optimizer and sampler-matched feature statistics.
Causal adaptation	Causal BestRQ	Strictly causal attention; unchanged targets.
Semantic supervision	Multi-task prediction	4096-unit multilingual CTC, Mel, and Chroma losses with weights 1/1/1.
Discretization	Vector quantization	16-dimensional cosine VQ, data-dependent initialization, exact all-rank usage statistics, and no dead-code revival.

The first phase learns broad music representations without lyric labels through BestRQ masked prediction: 0.4s waveform spans (40 frames) are masked with probability 0.3 and perturbed by -20 dB relative-RMS noise, and the four-frame target is quantized against a frozen 8192×16 random-projection codebook. Training uses deterministic 30-second crops from Dataset-Inventory under a source-balanced sampler. Domain adaptation then uses the annotated domain on 175,204 records (8,509 hours) from Dataset-Annotated and MiniMax with a fresh optimizer. Causal adaptation makes self-attention strictly causal so the encoder matches its autoregressive downstream use, using a global active-frame micro-average; targets are unchanged.

Semantic supervision first trains a new 4,096-unit multilingual subword CTC head alone, then thaws the backbone with CTC, Mel, and Chroma losses at equal weights (signed L1 for Mel and Chroma), using complete tracks up to 300 seconds and subwords rather than characters to respect the 25 Hz CTC capacity. This use of recognition supervision at a discrete bottleneck is closely related to ASR-guided speech content quantization, where the bottleneck is trained to preserve linguistic information (Liu et al., 2025). The final phase inserts the cosine quantizer (code dimension 16, commitment $\beta = 0.25$, EMA codebook decay 0.99) and trains with CTC/Mel/Chroma/VQ weights 0.5/1/1/1. The codebook is initialized from 1,048,576 real frames via PCA whitening and spherical k -means; a gate warmup and a frozen-lower phase ease the continuous-to-discrete transition, using exact all-rank usage statistics with no dead-code revival.

3.2 Music LLM and Melody-CoT

Reconstruction scope. Qwen-Music autoregressively predicts Semantic Tokens from musical tags and structured lyrics, optionally after generating a Melody-CoT plan (Xu et al., 2026). Its reported 3B Qwen3.5-Omni initialization is not publicly available. We use the public Qwen2.5-Omni-3B thinker (Xu et al., 2025a), preserving model scale but not architecture. A deterministic serializer replaces the unavailable Prompt Rewriter. Owing to limited data and compute, our LLM is pretrained only on a fixed English corpus of section-based windows no longer than 30 seconds, without curriculum learning. No post-training is performed—no supervised fine-tuning, offline DPO, or online GSPO—unlike Qwen-Music, which reports a full multi-stage post-training pipeline.

Sequences and token registry. Each record contains structured music tags, ordered song sections, lyrics, Semantic Tokens, optional section-aligned Melody Tokens, and provenance. We implement the three published layouts:

$$\begin{aligned}
\mathbf{s}_{\text{plain}} &= [\mathbf{x}, \mathbf{z}], \\
\mathbf{s}_{\text{section}} &= [\mathbf{x}, \mathbf{m}_{\text{section}}, \mathbf{z}], \\
\mathbf{s}_{\text{unique}} &= [\mathbf{x}, \mathbf{m}_{\text{unique}}, \mathbf{z}],
\end{aligned}$$

where $\mathbf{x}$ is the serialized text condition, $\mathbf{m}$ is Melody-CoT, and $\mathbf{z}$ is the Semantic Token sequence. Section mode includes one melody segment for every lyric-bearing section. Unique-section mode groups segments by section label and samples one representative per label anew in each epoch. The implementation retains all three layouts, including Melody-CoT, while the Music LLM finally uses the direct text-to-semantic layout (i.e., plain mode). The two Melody-CoT layouts remain available as alternative interfaces.

Token-level coupling between an internal plan and an acoustic response has also been explored in large speech models (Xie et al., 2025); our layouts instantiate the related planning-generation interface for music, while final inference uses the direct layout described above.

Melody Tokens follow the Qwen-Music rule. RMVPE first estimates pitch (Wei et al., 2023). After reduction to 50 Hz, eight-frame median pooling produces a 6.25 Hz sequence. Voiced frames encode a clipped MIDI offset from the median voiced pitch; token 255 denotes unvoiced frames.

A token registry assigns disjoint global ranges to text, control, Semantic, and Melody tokens; the vocabulary is padded to a fixed size. Render consumes local Semantic IDs in $[0, 32768)$, never the expanded global IDs.

Objective and context. Final training uses a 4,096-token per-GPU budget and at most 750 Semantic frames (30 seconds) per section-based example. The released inference configuration uses the same explicit Semantic-frame budget. We optimize next-token cross-entropy on Melody (optionally) and Semantic positions while masking the text condition, with no auxiliary loss or label smoothing.

Data and optimization. The training corpus contains 291,569 English windows. The windowing procedure follows annotated section boundaries and limits every example to 30 seconds. Its three quality strata are sampled with fixed weights 0.31479, 0.12719, and 0.55802.

Table 4: Optimization recipe for the autoregressive planner.

Setting	Music LLM
Initialization	Qwen2.5-Omni thinker
Learning rate / schedule	1×10^{-4} / constant
Parallelism	64 H20 GPUs, DDP, BF16
Context	4,096 tokens; ≤ 30 s / 750 Semantic frames

The planner is trained using DDP on 64 H20 GPUs, BF16 computation, AdamW at a constant 10^{-4} learning rate, gradient clipping at 2, and two-step gradient accumulation.

Decoding configuration. The decoder uses temperature 0.7, top- p 0.98, Semantic repetition penalty 0.1, and a 64-token repetition window.

3.3 Acoustic Render

Published design and scope. Qwen-Music Render turns frame-aligned Semantic Tokens and text conditions into audio: a 1.3B conditional Diffusion Transformer maps the conditions to a 128-dimensional, 25 Hz acoustic latent, a spectral VAE decodes the latent into a coarse 48 kHz stereo complex spectrogram, and a Band-Mode Refiner corrects frequency-band residuals before inverse STFT (Xu et al., 2026). The paper discloses the backbone geometry (32 blocks, 1024 hidden width, 24 attention heads, expansion 8, self-attention $\rightarrow$ cross-attention $\rightarrow$ feed-forward order, timestep AdaLN, RoPE), the text encoders (a frozen Qwen3-Embedding-0.6B model for descriptions, a six-layer transformer for lyrics, text-only dropout for CFG), the VAE interfaces ($[2, 480, T]$ complex spectra over a SpectroStream-style seven-block encoder/decoder with delayed-fusion stereo), and the Refiner’s outer training recipe. It omits the STFT configuration, posterior form, DiT head geometry, semantic fusion operator, flow path and timestep distribution, solver, CFG, and EMA settings, and most optimization budgets. The values below are our reconstruction decisions, not claims about the official implementation.

Training recipe. Table 5 lists the Render components and their training objectives. As with the tokenizer, every model artifact is bound by SHA-256 in the artifact manifest.

Table 5: Training recipe for the acoustic renderer.

Component	Objective	Main reconstruction choice
Spec-VAE	Reconstruction	Full VAE on 48 kHz stereo; global batch 256.
Spec-VAE	Decoder fine-tuning	Encoder frozen; STFT-domain adversarial training at batch 64.
DiT	Conditional flow matching	24-block, width-1536 backbone; exponential-moving-average weights.
Refiner	Band-aware residual correction	Frozen VAE with fresh optimizers; mid-band magnitude and phase correction.

Spec-VAE. Our Spec-VAE preserves the published interfaces. Audio enters at 48 kHz stereo through an STFT with $n_{\text{fft}} = 960$ and an equal window length, hop 480, no centering, and a 240-sample left pad; DC is retained and Nyquist dropped, giving 480 frequency bins per channel and the $[2, 480, T]$ representation. The encoder fuses stereo channels late and the decoder splits them early, both in the published seven-block 2D convolutional style. The posterior is a diagonal Gaussian: training draws posterior samples, while validation reconstruction uses the mean. Initial training fits the full VAE on Dataset-Inventory at batch 256. Decoder fine-tuning then freezes the encoder and uses batch 64 while adding an STFT-domain adversarial objective (multi-resolution STFT, a mixed-scale spectral term, and waveform adversarial and feature-matching terms).

Conditional DiT. The renderer uses a Stable Audio 3-compatible profile: 24 blocks, hidden width 1536, 24 attention heads of dimension 64, feed-forward multiplier 4, QK RMS normalization with differential attention, and 64 learned memory tokens. Local, global, and text conditioning each use width 1024; Semantic Tokens remain aligned at 25 Hz to the 128-dimensional spectral latent.

The model is trained by conditional flow matching with the VAE encoder frozen.

Training uses 64 H20 GPUs, batch one per rank, four-step gradient accumulation (global parent batch 256), BF16, AdamW at 5×10^{-5} with betas 0.9/0.95 and weight decay 0.01, a brief warmup followed by a constant rate, gradient clipping at 1.0, and FP32 exponential moving averages with decay 0.999. A global loudness condition uses -15.4335 LUFS .

EMDC. Because the Music LLM generates rather than copies Semantic Tokens, the DiT’s training conditions are cleaner than those it sees in deployment. We adapt the EMDC strategy of FullDiT (Dai et al., 2026): after clean training, each semantic frame is independently replaced with probability p_{replace} by a code sampled from the K cosine-nearest neighbors of its ID in the tokenizer’s 16-dimensional codebook. The replacement probability, neighborhood size, and schedule are configuration parameters recorded in the configuration files; optimizer, moving-average policy, data, and VAE remain unchanged during this adaptation.

Band-Mode Refiner. The Refiner sits between the spectral decoder and inverse STFT, using a 12-layer ConvNeXt-1D trunk (width 256, kernel 7) with hard band boundaries at 1.5 and 4 kHz: low band phase only, mid band magnitude and phase, high band magnitude only. Output projections are zero-initialized so the untrained network is a learnable identity (our interpretation of the published zero initialization), and phase errors use wrapped-L1 instantaneous-frequency and group-delay terms. Refiner training freezes the VAE and uses fresh optimizers under the published outer recipe.

Inference configuration. The inference pipeline chains the diffusion renderer, spectral VAE, and band-aware Refiner, with an 8-step Euler solver, CFG 1.0, and global loudness -15.4335 LUFS. The artifact manifest binds these artifacts and their configurations.

4 Evaluation

We report held-out quantitative evaluations of automatic supervision and semantic discretization, followed by a qualitative listening assessment of the complete generation pipeline. Each quantitative table states its evaluation set, sample count, aggregation, and metric scale.

4.1 Annotation Quality

The training pipeline relies on automatically generated lyrics and musical attributes. Table 6 reports the measurements for which the audit protocol is fixed.

Table 6: Automatic-annotation audit. Each row retains its own task-specific denominator and is not combined into an overall annotation score.

Signal	Result	Note
Lyric LCS recall	0.889	precision 0.879; CI [0.877, 0.902]
Vocal gender	88.3%	CI [85.9, 90.3]
Genre	75.2%	GTZAN subset
Mood	76.9%	arousal after dead-zone mapping
Confidence–error correlation	Spearman -0.764	two-transcription confidence

The negative confidence–error correlation indicates that agreement between the two transcription paths is informative for data-quality filtering. Because the rows measure different annotation tasks and use different reference subsets, we interpret them individually rather than averaging them.

4.2 Semantic Tokenizer Quantization

We measure the information loss introduced by assigning continuous features to discrete codes. The same examples and fixed prediction heads are applied to the 16-dimensional projected representation (P) and its nearest-code quantized counterpart (Q). Table 7 reports the increase in prediction error from P to Q ; lower values indicate better preservation.

Table 7: Quantized-minus-continuous ($Q - P$) prediction-error differences. PER differences are percentage points. Mel was not measured on the common music holdout.

Evaluation panel	Examples	Δ PER (pp)	Δ Mel	Δ Chroma
Common music holdout	2,372	2.352	—	0.00124
Exposure panel	256	2.153	0.0306	0.00159
Real-recording panel	101	2.879	0.0267	0.00143

Approximately 98.2% of the 32,768-entry codebook is active across the evaluated panels, indicating no evident codebook collapse on these data. Qwen-Music reports more than 99% utilization under its own protocol (Xu et al., 2026); because utilization depends on the corpus and the number of observed frames, the values provide context rather than a controlled ranking.

The 2.15–2.88 percentage-point PER increase quantifies the additional phonetic prediction error associated with nearest-code assignment. Mel and Chroma differences provide complementary within-system diagnostics. No external result uses the same paired $P \rightarrow Q$ protocol, so these measurements are not used for cross-system ranking.

4.3 Qualitative Listening Observations

We qualitatively assess two paths through the frozen generation pipeline. The demos are shown at <https://huggingface.co/spaces/oqmtest1451/open-qwen-music-demo>. This assessment records recurring listening observations rather than numerical preference estimates.

Reconstruction from reference-audio Semantic Tokens. Semantic Tokens extracted from a source recording are passed directly to the acoustic renderer. The resulting audio generally preserves the sung lyrics: most words remain intelligible and follow the source content. The main audible limitations are an electronic coloration in the vocal timbre and a stereo image with limited width and depth.

End-to-end text-to-music generation. The Music LLM first generates Semantic Tokens from the text and lyric conditions, after which the same acoustic renderer synthesizes the waveform. In this setting, some lyric words or phrases are omitted, while others are rendered indistinctly. Relative to reconstruction from reference-audio Semantic Tokens, this pattern suggests that errors in the Music-LLM-generated semantic sequence are an important contributor to the additional degradation observed in the rendered output.

5 Discussion

Summary. Open-Qwen-Music reconstructs the Qwen-Music semantic-to-waveform pipeline under a constrained but fully specified setting: approximately 10,100 hours of training exposure compared with the more than five million hours reported by Qwen-Music, one 64-GPU budget per module, automatically generated annotations, and public replacements for unavailable components (Xu et al., 2026; MiniMax AI, 2026; Xu et al., 2025a). Within these constraints, the system completes the full path from text and lyrics to 48 kHz stereo audio. The purpose of the reconstruction is to make the architecture reproducible and inspectable rather than to claim quality parity with Qwen-Music. We therefore release the training resources, model weights, and data-processing, training, inference, and evaluation pipelines needed to reproduce the system.

Perceptual characterization. Qualitative listening separates two sources of degradation. When the renderer is conditioned on reference-audio Semantic Tokens, most lyric content remains intelligible, showing that the acoustic stack can express vocal content from a reliable semantic sequence. The reconstructed vocals nevertheless retain an electronic coloration, and the stereo image has limited width and depth. In end-to-end text-to-music generation, where the Music LLM supplies the Semantic Tokens, generation quality is less stable and vocals are frequently missing or indistinct. This contrast suggests that the quality of Music-LLM-generated Semantic Tokens is an important contributor to failures in vocal intelligibility. The renderer also limits timbral and spatial fidelity, and the present qualitative comparison does not isolate the relative causal contributions of the two stages.

Main problems. The current system has several limitations.

- *Automatic supervision remains imperfect.* The annotation audit reports lyric LCS recall of 0.889 and attribute accuracies between 75.2% and 88.3%. Errors in lyrics and musical attributes can propagate into the tokenizer, Music LLM, and renderer conditions.
- *Semantic discretization introduces measurable information loss.* The codebook is broadly active, with 98.2% utilization on the evaluated panels, while nearest-code assignment raises PER by 2.15–2.88 percentage points relative to the projected continuous representation. The projected features are normalized before cosine quantization. This design may improve codebook coverage while discarding magnitude-related information, and it may increase the difficulty of downstream sequence modeling. Establishing this causal link requires additional evidence, such as conditional-entropy estimates, Music-LLM negative log-likelihood, or a controlled comparison between tokenizer variants.
- *Music-LLM-generated Semantic Tokens are an important contributor to vocal degradation.* The model is trained on English section windows no longer than 30 seconds. In qualitative end-to-end

generation, its Semantic Tokens lead to unstable output and frequent missing or indistinct vocals, while an independent held-out Music LLM evaluation is not yet available.

- *Acoustic rendering limits fidelity.* With reference-audio Semantic Tokens, the renderer generally preserves lyrics but retains electronic vocal coloration and weak stereo width and depth.
- *Evaluation coverage remains limited.* Current quantitative results cover annotation quality and tokenizer quantization. The system-level listening observations are qualitative, and no matched comparison with another text-to-music system is reported.

Planned improvements. These limitations define the next iteration of the project.

- *Data.* Expand the training corpora while preserving item-level provenance and redistribution controls, improve the automatic annotation models, and introduce human-verified subsets for annotation calibration.
- *Representation.* Reduce the information lost at the semantic bottleneck and evaluate tokenizer changes by their effect on downstream generation as well as by component-level diagnostics.
- *Music LLM.* Improve the stability and completeness of generated Semantic Tokens, then evaluate the model on an independent held-out set with lyric coverage, termination, and semantic consistency measures. Evaluation will also extend to additional languages and longer cross-section contexts.
- *Post-training.* Curate high-quality supervision and investigate multi-objective alignment across musicality, lyric intelligibility, and control adherence, using rewards computed from rendered audio rather than token sequences alone. This direction is motivated by both multi-objective LLM alignment and perceptual-feedback optimization in generative speech, where token-level objectives need not track perceptual quality (Li et al., 2025b;a).
- *Rendering.* Reduce electronic vocal coloration, improve stereo width and depth, and strengthen robustness to Music-LLM-generated Semantic Tokens while retaining the lyric preservation observed with reference-audio Semantic Tokens.
- *Evaluation.* Add multi-seed generation, controlled human listening, and protocol-compatible comparisons with public text-to-music systems.

Open Release and Outlook. The remaining gap is not a missing pipeline component; it lies in data scale and supervision quality, in the Semantic Token representation formed by the tokenizer, in the acoustic and spatial fidelity of the renderer, and in the current evaluation coverage. In particular, normalization and quantization discard part of the input information and may increase the difficulty of modeling the resulting Semantic Token sequences; this hypothesis requires controlled tokenizer comparisons or direct sequence-modeling measurements. Although the present results do not establish quality parity with Qwen-Music, we release the system in full. To our knowledge, Open-Qwen-Music is the first fully open release of a text-to-music system that combines LLM-based semantic composition with diffusion-based acoustic rendering and makes every layer of the research stack public: training datasets with provenance manifests, data-processing and annotation pipelines, model definitions, training and inference code, evaluation code and configurations, and pretrained weights for every learned module. We intend it as a reproducible foundation for research on this architecture and broader audio-language modeling (Xie et al., 2026a), and will continue to improve its generation quality, controllability, and robustness.

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A Detailed Data Information

A.1 Dataset-Inventory Source Breakdown

Table 8: The 26 retained sources in Dataset-Inventory. Hours are rounded to two decimals. Each row cites the primary paper, the ingested release page, or both. Derivative and mirror caveats are stated below.

Source	Records	Hours
AI-Music Deduplicated (AI-Music, 2026)	508,557	25,141.23
JamendoMaxCaps (Roy et al., 2025)	261,739	13,683.13
Muse (Jiang et al., 2026)	117,067	7,589.09
Udio Dataset (Blanchon, 2024)	134,329	3,914.82
FMA Full (Defferrard et al., 2017)	65,282	3,607.52
GlobalDISCO (Solak et al., 2025)	73,024	3,340.90
MTG-Jamendo Raw 30s (Bogdanov et al., 2019)	45,342	2,636.36
ACE-Step Songs (Yi3852, 2025; Jiang et al., 2025)	21,568	1,012.61
Music4All (Santana et al., 2020)	95,699	797.49
Cambridge-MT (Senior, n.d.)	6,856	409.86
Jamendo-QA mirror (Archit00, 2026; Koh et al., 2025)	6,396	394.31
MuChin-v2 (karl-wang, 2025; Wang et al., 2024)	4,216	269.85
AnyInstruct Music (Zhan et al., 2024)	118,368	166.37
MoisesDB (Pereira et al., 2023)	1,348	79.00
OpenMIC-2018 (Humphrey et al., 2018)	16,789	46.64
MAESTRO v3.0.0 (Hawthorne et al., 2019)	367	24.10
MERGE Bimodal (Louro et al., 2024; 2026)	1,570	13.07
MusicNet (Thickstun et al., 2017)	130	8.92
MUSDB18-HQ (Rafii et al., 2019)	140	8.59
PMemo2019 (Zhang et al., 2019; 2018)	780	8.24
MedleyDB-derived mix (AMAAI Lab, n.d.; Bittner et al., 2014; 2016)	129	6.78
Coimbra MIREX (Panda et al., 2013)	726	5.98
GuitarSet (Xi et al., 2018)	360	3.05
URMP (Li et al., 2019)	42	1.28
MusicCaps raw-audio mirror (Google Research, 2023; Agostinelli et al., 2023)	253	0.70
Song Generation Binisha (binisha, 2024)	224	0.51

The source labels identify the artifacts ingested by our pipeline, not necessarily the official upstream distributions. In particular, Jamendo-QA is the `Archit00` mirror. The PMemo row uses the updated 2019 release. The MedleyDB row is an AMAAI Lab derivative containing instrument/other mixes, rather than the official multitracks. MusicCaps raw audio comes from a local third-party mirror containing 326 candidates before filtering, whereas the official MusicCaps release contains 5,521 metadata records. Its citations identify the upstream benchmark, not an independently archived provenance record for that audio mirror.

A.2 Dataset-Annotation-Filtered Eligibility Rules

Table 9: Main Dataset-Annotation-Filtered eligibility rules. The implementation also applies silence, timeline, and annotation-consistency checks.

Branch	Required evidence	Main numeric rules
Vocal	Melody-CoT eligible; at least two sections; no under-transcribed, dropped, or hallucinated lyric windows; English, Chinese, or bilingual.	Lyric confidence ≥ 0.905 ; lyric coverage ≥ 0.15 ; genre ≥ 0.80 ; instrument ≥ 0.75 ; binary vocal gender ≥ 0.89 .
Instrumental	Non-synthetic; no usable lyrics or vocal tags; fewer than three text units.	Instrumental-gender confidence ≥ 0.90 ; vocal ratio < 0.15 ; vocal-stem-to-mix ratio < -15 dB.

Lyric confidence thresholds for broader task eligibility are 0.723 for pretraining, 0.905 for Melody-CoT, and 0.945 for SFT. Tags are retained at confidence 0.40 or above. These are fixed filtering thresholds rather than estimates of human-label accuracy; the SFT threshold is reserved for future supervised fine-tuning data curation.

A.3 Annotation Pipeline

The released annotation pipeline is a dependency-aware directed acyclic graph with 12 stages. Independent evidence streams are preserved until fusion: the two transcription passes are produced separately, and generative tag predictions are retained separately from acoustic voice evidence. Table 10 summarizes the input-output role of each stage.

Table 10: Stages of the released annotation pipeline. The order follows the dependency graph; stages whose prerequisites are satisfied may run in parallel.

Stage	Function
1 Index	Ingest source manifests or file trees, assign stable sample identities, and record source and audio metadata.
2 Source separation	Extract the vocal stem and vocal-activity evidence used by the lyric and voice-analysis branches.
3 Structure analysis	Detect musical-section boundaries and normalize predicted section labels to the shared section taxonomy.
4 Vocal-stem transcription	Produce the primary lyric transcription from the separated vocal stem.
5 Mixture transcription	Independently transcribe the original mixture for cross-checking the vocal-stem result.
6 Lyric fusion	Reconcile the two transcription streams with available source text, determine language, and produce the selected lyric sequence with quality measurements.
7 Lyric alignment	Assign word- or character-level timestamps to the selected lyric sequence.
8 Section assembly	Combine structural boundaries, aligned lyrics, and vocal activity into section-level records and structured lyrics.
9 Voice acoustics	Extract pitch and spectral evidence from the vocal stem for the vocal attributes used during tag fusion.
10 Generative tagging	Produce a free-form description and candidate genre, mood, instrument, vocal-gender, and vocal-timbre tags under controlled output taxonomies.
11 Tag fusion	Normalize tag candidates and fuse generative, acoustic, and structural evidence while retaining per-attribute confidence and source information.
12 Validation and export	Assemble the final annotation record, apply confidence gates and downstream-use eligibility rules, validate the schema, and emit the training manifest.

A.4 Annotation Output Schema

Each exported row follows a single versioned record schema. Downstream corpus construction reads this record rather than stage-specific intermediate files. Table 11 lists the public field groups and their semantics.

Table 11: Field groups in each exported annotation record.

Field group	Contents	Role
Identity	Schema version, stable sample identifier, and source-dataset label.	Joins annotations to the corresponding corpus item while preserving the record contract.
Audio	Audio reference, duration in seconds, sample rate, and channel count.	Defines the media item and its technical properties.
Language	Resolved lyric language, when available.	Supports language-aware normalization, filtering, and corpus sampling.
Description	Free-form musical description produced by the tagging branch.	Provides text conditioning beyond the controlled tag vocabulary.
Musical tags	Genre, mood, instrument, vocal gender, and vocal timbre; low-confidence attributes are omitted.	Supplies independently controllable prompt attributes without treating missing evidence as a negative label.
Structured lyrics	Lyrics serialized in section order with section labels.	Provides the text and musical-form representation consumed by downstream sequence construction.
Sections	Section identifier and label, start and end time, lyric and vocal flags, section text, aligned-unit count, and lyric coverage.	Preserves the time-aligned relationship among form, vocals, and lyrics.
Provenance and quality	Lyric-source status and confidence, alignment availability, tag confidence and evidence sources, and processing-completeness indicators.	Makes filtering decisions traceable without requiring intermediate-stage outputs.
Usability	Eligibility flags and reasons for supported downstream training uses.	Allows each training corpus to apply its own evidence requirements from the same annotation record.

B Training and Inference Configuration

This section consolidates the optimization and decoding settings used for the reported system. The tables separate the sequential training phases and report effective global batch sizes after gradient accumulation.

B.1 Semantic Tokenizer

All six optimization phases use 64 GPU processes, one gradient-accumulation step, BF16 computation, and gradient clipping at 1.0. AdamW uses $\beta_1 = 0.9$ and $\beta_2 = 0.98$; the learning rate is linearly warmed up and then cosine-decayed to one tenth of its peak value. Weight decay is 0.01 except during supervised-head warmup, where it is zero.

Table 12: Sequential optimization phases for the semantic tokenizer. Batch/GPU is the per-process batch size; warmup and updates count optimizer steps.

Phase	Objective and training unit	Updates	Batch/GPU	Global batch	Peak LR / warmup
Masked pretraining	Bidirectional BestRQ on deterministic 30-s crops	100,000	2	128	2×10^{-4} / 10,000
Music-domain adaptation	BestRQ on deterministic 30-s crops from the annotated domain	20,000	4	256	2×10^{-5} / 500
Causal adaptation	Causal BestRQ with unchanged acoustic targets	20,000	4	256	2×10^{-5} / 500
Supervised-head warmup	Frozen backbone; CTC head on complete tracks up to 300 s	2,000	1	64	1×10^{-3} / 100
Joint semantic supervision	CTC, Mel, and Chroma losses with weights 1/1/1	40,000	1	64	5×10^{-5} / 3,000
Discrete codebook training	CTC/Mel/Chroma/VQ weights 0.5/1/1/1	25,000	1	64	3×10^{-5} / 2,000

Masked pretraining uses 0.4-s waveform spans with masking probability 0.3 and -20 -dB relative-RMS noise. The final three phases use a 4,096-unit multilingual CTC vocabulary; Mel and Chroma use signed L1 losses. Before the last phase, the $32,768 \times 16$ cosine codebook is initialized from 1,048,576 projected frames by PCA whitening and five iterations of mini-batch spherical k -means. The quantizer uses commitment weight 0.25, a 5,000-step gate warmup, and a 5,000-step frozen-lower-encoder interval.

B.2 Music LLM

The Music LLM is optimized only in the direct text-to-semantic sequence layout. Both stages use 64 GPU processes, one sequence per process, two gradient-accumulation steps, BF16 computation, and an effective global batch size of 128. Stage 2 continues the data cursor from Stage 1 but starts a new optimizer.

Table 13: Two-stage Music LLM optimization. Cumulative updates count the uninterrupted sequence of consumed training batches.

Stage	Initialization and data progression	Stage updates	Cumulative updates	Global batch
Stage 1	Qwen2.5-Omni-3B initialization; beginning of the corpus stream	5,000	5,000	128
Stage 2	Stage-1 model initialization; continuation of the corpus stream	5,000	10,000	128

Both stages use AdamW with learning rate 1×10^{-4} , betas 0.9/0.95, weight decay 0.01, a constant schedule without warmup, and gradient clipping at 2.0. The per-GPU token budget is 4,096, with at most 750 Semantic frames per example. Training uses 291,569 English windows of at most 30 s, sampled from the three data strata with probabilities 0.31479, 0.12719, and 0.55802.

B.3 Acoustic Models

All acoustic modules operate on 48-kHz stereo audio. The Spec-VAE and Refiner use 1.28-s random crops; the Renderer predicts 128-dimensional VAE latents at 25 Hz. Table 14 gives the effective distributed schedules.

Table 14: Distributed training schedules for the acoustic modules.

Module or stage	Objective	Updates	GPUs	Batch/GPU	Accum.	Global batch	Precision
VAE Stage 1	Full-model reconstruction and KL training	90,000	64	4	1	256	FP32
VAE Stage 2	Decoder-only adversarial refinement	5,000	64	1	1	64	FP32
Renderer	Conditional flow matching	29,000	64	1	4	256	BF16
Refiner	Band-aware residual correction	1,250	64	1	1	64	FP32

Spec-VAE. The common analysis STFT uses a 960-sample Hann window, a 480-sample hop, no centering, and a 240-sample left pad; DC is retained and Nyquist is removed. Stage 1 uses Muon at 1×10^{-4} with momentum 0.95, weight decay 0.1, and a constant learning rate. Its objective combines multi-resolution STFT loss (weight 1), KL regularization (10^{-6}), and mixed-scale spectral loss (2×10^{-6}). Stage 2 freezes the encoder and uses separate Muon optimizers for the decoder and discriminator at 3.75×10^{-5} and 3.75×10^{-6} , respectively. Both use momentum 0.95; the learning rate warms up for 1,000 steps and cosine-decays to 10% of its peak. Stage 2 adds waveform adversarial and feature-matching losses with weights 0.03 and 3.0.

Renderer. The Renderer uses linear rectified flow, a Gaussian source, uniformly sampled timesteps, velocity prediction, and uniform loss weighting. AdamW uses learning rate 5×10^{-5} , betas 0.9/0.95, weight decay 0.01, a 2,000-step warmup followed by a constant rate, and gradient clipping at 1.0. FP32 exponential-moving-average parameters use decay 0.999. The first 25,000 updates use clean Semantic Tokens, followed by a 1,000-update corruption ramp; the remaining updates use the full mixture, in which 25% of examples receive codebook-neighbor corruption with $K = 384$.

Refiner. The Refiner keeps the VAE fixed and alternates four generator updates per discriminator update. AdamW learning rates are 1.5×10^{-4} for the generator, 1×10^{-6} for the waveform discriminator, and 1×10^{-5} for the spectral discriminator; all use betas 0.9/0.95 and weight decay 0.01. The objective weights are 0.5 for STFT reconstruction, 0.1 for instantaneous-frequency/group-delay reconstruction, 0.175 and 20 for waveform adversarial and feature matching, and 0.5 and 35 for spectral adversarial and feature matching.

B.4 Inference

Table 15: Inference settings used for text-to-music generation.

Setting	Value
Music LLM sequence layout	Direct text-to-semantic
Sampling	temperature 0.7; top- p 0.98; top- k 0
Semantic repetition control	penalty 0.1 over a 64-token window
Acoustic solver	Euler, 8 steps
Classifier-free guidance	1.0
Loudness condition	-15.4335 LUFS
Output	48-kHz stereo waveform